INDEX

M1 – Method mark, awarded for correct mathematical technique, independent of numbers used.

A1 – Answer mark, awarded for a correct result to a calculation, dependent on M

B1 – Independent mark, can be awarded regardless of other marks, for a correct method

E1 – Explanation mark, for correct use of terminology and mathematics for justification

F – Follow through from an incorrect result

AWRT – Answers which round to a given number

AWFW – Answers which fall within a given range

ACF – Any correct form, answers which are mathematically equivalent to the given answer

OE – Or Equivalent, numbers expressed in different forms

NE – not enough, not awardable/insufficient

A candidate may only lose a maximum of one mark throughout the paper for omission of units

Q	Marking Instructions	Marks	Notes
1 (i)	(A) The distribution has a negative skew (B) As r gets larger $\left(r-\frac{1}{r}\right)$ also grows larger, making higher values more likely, hence a negative skew	**B1** **E1**	
	Sub Total	**2**	
1 (ii)	$\frac{3k}{2}+\frac{8k}{3}+\frac{15k}{4}+\frac{24k}{5}=1$ $\frac{763k}{60}=1 \Rightarrow k=\frac{60}{763}$	**B1** **B1**	**For at least one probability** **For a correct method to evaluate k**
	Sub Total	**2**	
1 (iii)	$k=\frac{60}{763}$ $3k+8k+15k+24k=\mathrm{E}(X)$ $\mathrm{E}(X)=3.93\ldots(184797)$ $\mathrm{Var}(X)=1.05\ldots(433302)$	**M1** **A1** **A1**	**Either formula seen** **Answers to 3 s.f or better**
	Sub Total	**3**	
	Question Total	**7**	

Q	Marking Instructions	Marks	Notes
2 (i)	The data collected on the scatter diagram can be evenly spread within a roughly **elliptical** shape, suggesting the underlying parent population is likely a **bivariate normal distribution,** making a test based on the PMCC suitable	**E1** **E1dep**	 **Not necessary to mention population**
	Sub Total	**2**	
2 (ii)	H_0: There does not exist a correlation between a further maths student's statistics ability and discrete maths ability **in the population** H_1: There exists some correlation between a further maths student's statistics ability and discrete maths ability **in the population** For a 2 tailed test where n=30, the critical value at 5% is 0.3610 $0.511 > 0.361$ ∴ Reject H_0 There is sufficient evidence to suggest that there exists a correlation between a further maths student's statistics ability and discrete maths ability **in the population**	**B1** **A1** **B1** **M1** **A1**	**Must reference population** **Context, must reference population**
	Sub Total	**5**	
2 (iii)	A sample size of 30 is quite large, which leads to a smaller critical value at most significance levels, which means the null hypothesis is still likely to be rejected (even at smaller significance levels)	**E1**	**Comments on the size of the sample**
	Sub Total	**1**	
	Question Total	**8**	

Q	Marking Instructions	Marks	Notes
3 (i)	(A) $\mathrm{E}(X) = 0.162, \mathrm{Var}(X) = 0.1616$ Mean and variance are very close, making a Poisson distribution suitable (B) Fish must be sold at a **constant (average) rate,** and must occur **randomly** (and independently) of one another	**E1** **B1** **B1**	**Accept n is large and p is small** **Accept singular/one at a time**
	Sub Total	**3**	
3 (ii)	(A) $540 \times 60 \times 0.0027 = 87.48$ (B) $N \sim \mathrm{Po}(87.48)$ $p \approx 0.3674$	**A1** **A1**	 **f.t**
	Sub Total	**2**	
3 (iii)	(A) $P(N = 80) \times P(Y = 70) \approx 9.98 \times 10^{-4}$ (B) $Z \sim \mathrm{Po}(165.48)$ $\sigma \approx 12.864$ $P(152.6 \leq Z \leq 178.3)$ ≈ 0.6879	**M1** **A1** **M1** **M1** **A1**	 **f.t** **f.t**
	Sub Total	**5**	
	Question Total	**10**	

Q	Marking Instructions	Marks	Notes
4 (i)	The probability of the dog barking up the wrong tree remains **fixed**, and is **independent** of the previous observation's outcome	**E1** **E1**	
	Sub Total	**2**	
4 (ii)	The expected frequency of 0 or 1 was likely to be less than 5, making a χ^2 test unreliable	**E1**	
	Sub Total	**1**	
4 (iii)	H_0: Po(3.6) is a good fit for the data H_1: Po(3.6) is not a good fit for the data 	Value	≤1	2	3	4	5	≥6
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Observed	12	18	21	18	14	17
Probability	0.1257	0.1771	0.2125	0.1912	0.1377	0.1559
Expected	12.57	17.71	21.25	19.12	13.77	15.59
Contribution	0.0258	0.0047	0.0029	0.0656	0.0038	0.1275 | $\chi^2 = \sum contributions = 0.2303$ ν =columns – 2= 4 (due to estimated parameter) For a χ^2test, 5%, $\nu = 4$, c.v $= 9.488$ $0.2303 < 9.488$ not significant ∴ fail to reject H_0 There is insufficient evidence to suggest that a Poisson distribution with a mean equal to the mean of $B(300,0.012)$ is not a good fit to represent SCE's observations of an old dog barking up the wrong tree.	**B1** **M1** **A1** **A1** **A1** **B1** **B1** **A1**	**Hypotheses clearly stated** **One Probability calculated** **Expected frequencies** **Contributions** χ^2 **Correct c.v** **Comparison** **Conclusion in context**
	Sub Total	**8**	
	Question Total	**11**	

Q	Marking Instructions	Marks	Notes
5 (i)	Height is a controlled variable	**E1**	**Reference to height not being random**
	Sub Total	**1**	
5 (ii)	$\bar{d} = 1.018, \bar{h} = 1000$ $\frac{S_{hd}}{S_{hh}} = \frac{\left(7930 - \frac{9.16 \times 9000}{9}\right)}{\left(1.275 \times 10^7 - \frac{9000^2}{9}\right)}$ Gradient $= -3.28 \times 10^{-4}$ $d = 1.346 - 0.000328h$ For 490m, prediction $= 1.19$ For 2700m, prediction $= 0.460$	**M1** **A1** **A1** **A1**	**Correct equation (ACF)** **3s.f maximum**
	Sub Total	**4**	
5 (iii)	The prediction for $490m$ is likely to be reliable as it involves **interpolation.** The prediction for $2700m$ is unlikely to be reliable as it involves **extrapolation**	**E1** **E1**	
	Sub Total	**2**	
5 (iv)	Predicted value for 1750 = 0.772 Residual = 0.238	**M1** **A1**	
	Sub Total	**2**	
5 (v)	$r^2 = 0.992$ A large coefficient of determination shows that a change in height strongly suggests a change in air density, hence the students claim is correct	**M1** **E1 dep**	
	Sub Total	**2**	
	Question Total	**11**	

Q	Marking Instructions	Marks	Notes
6 (i)	(A) $\frac{1}{2}$ (B) $\frac{7n-3}{2}$ is the median which is a value in the event n is odd There are $\frac{n-1}{2}$ values below the median Hence the probability is $\frac{n-1}{2n}$	**B1** **E1** **M1** **A1**	**oe**
	Sub Total	**4**	
6 (ii)	Let U be the uniform distribution across $\{1,2,\ldots,n-1,n\}$ $X = 7U - 5$ $E(X) = \frac{7n-3}{2}$ $Var(X) = \frac{49n^2-49}{12}$	**M1** **M1** **A1** **A1**	**o.e (Accept recognition that E(X)=median)** **o.e**
	Sub Total	**4**	
6 (iii)	(A) $N \sim Geo\left(\frac{1}{n}\right)$ (B) $P(N \geq 3) = P(N > 2) = \left(1-\frac{1}{n}\right)^2$ $\left(1-\frac{1}{n}\right)^2 = \frac{81}{100}$ $\left(1-\frac{1}{n}\right) = \frac{9}{10}$ $n = 10$ (C) 90	**B1** **M1** **M1** **A1** **B1**	
	Sub Total	**5**	
	Question Total	**13**	